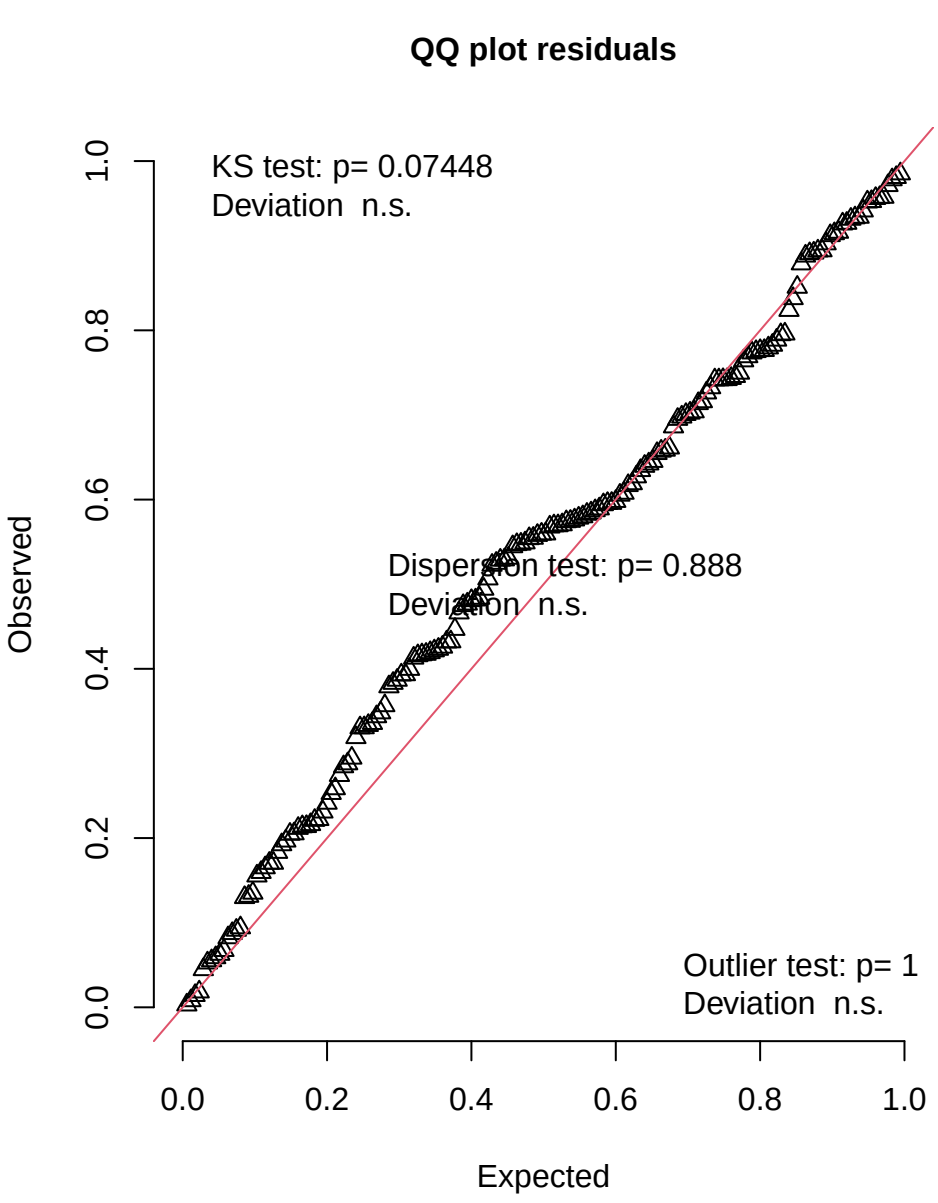
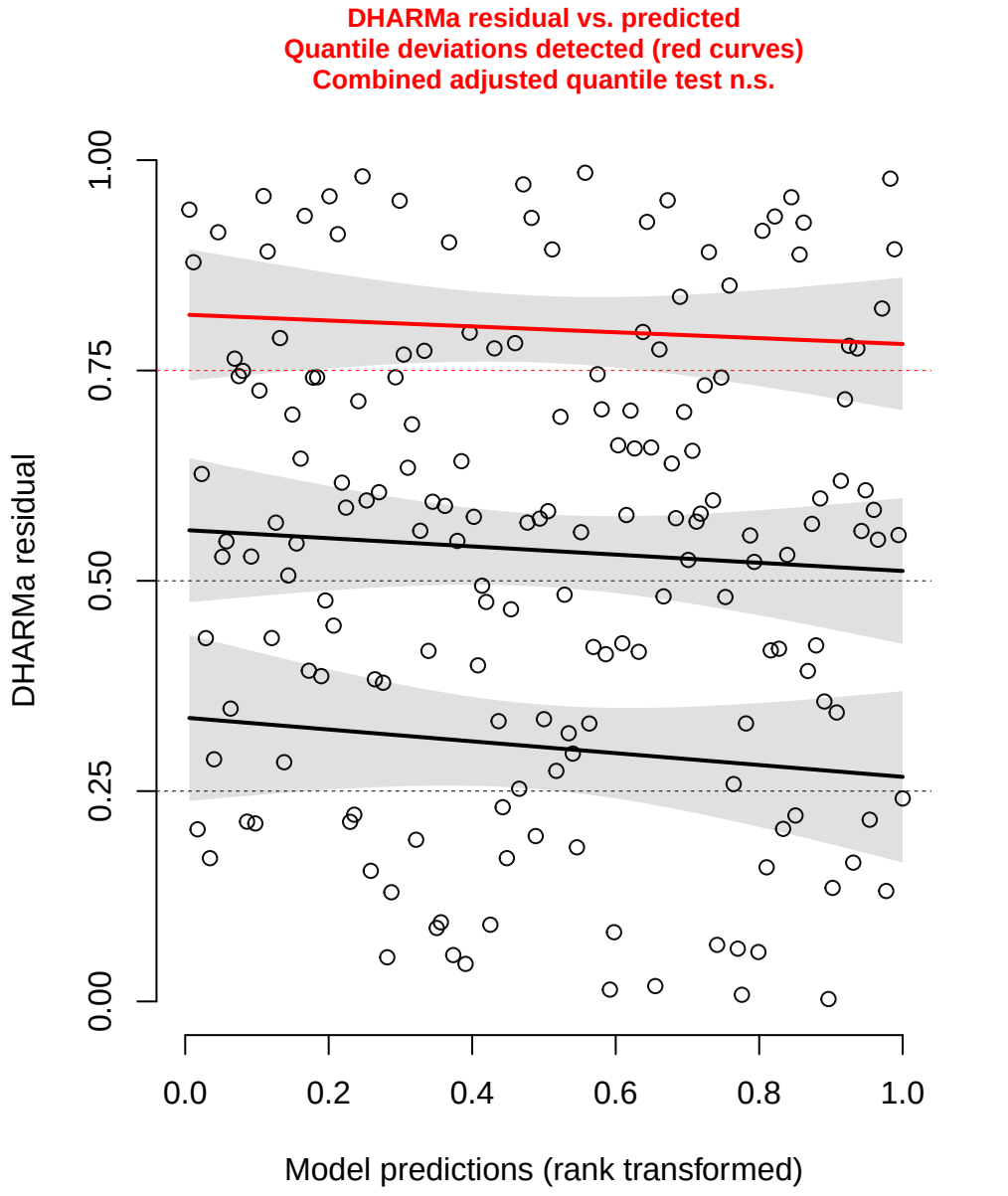
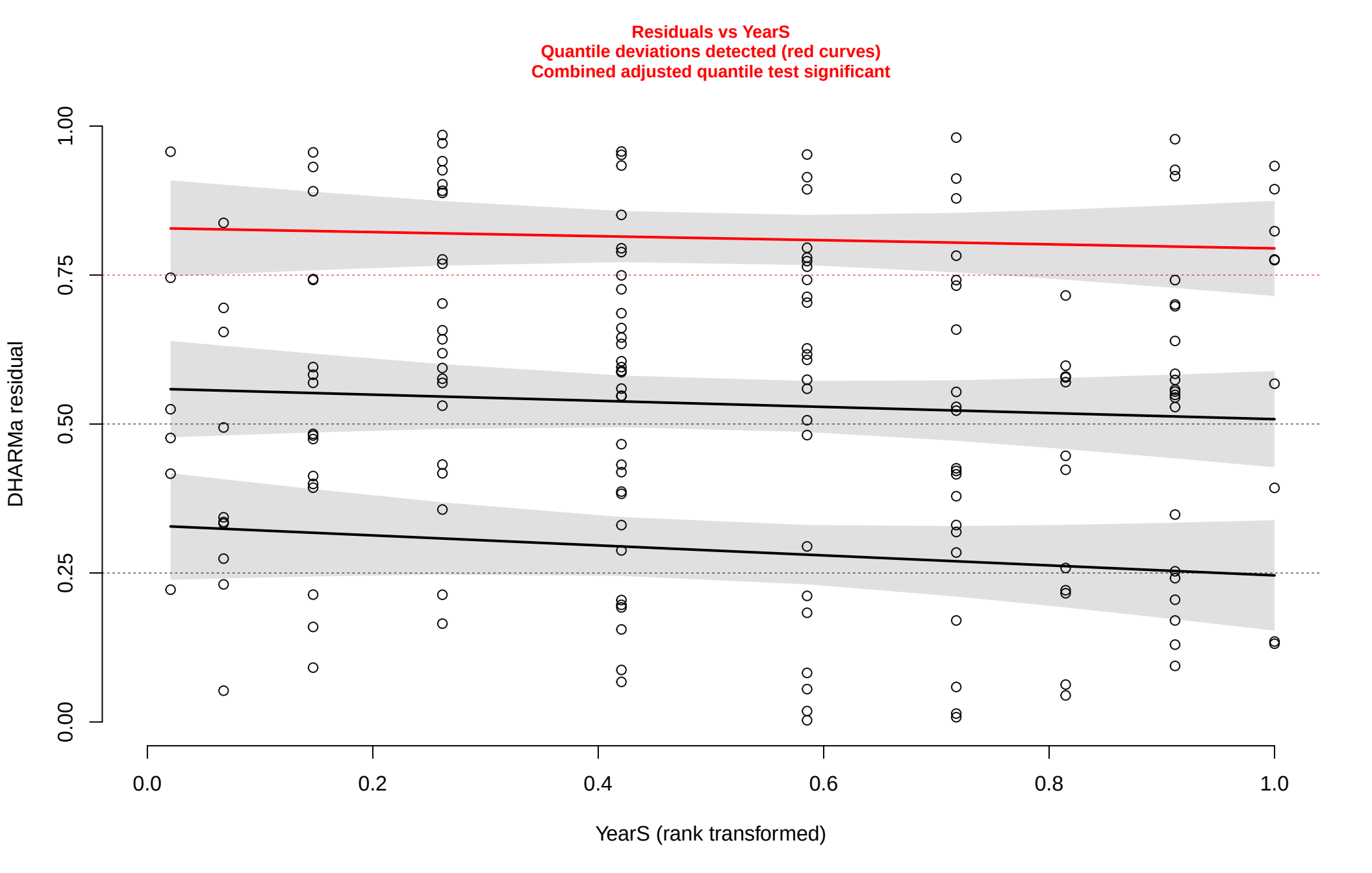
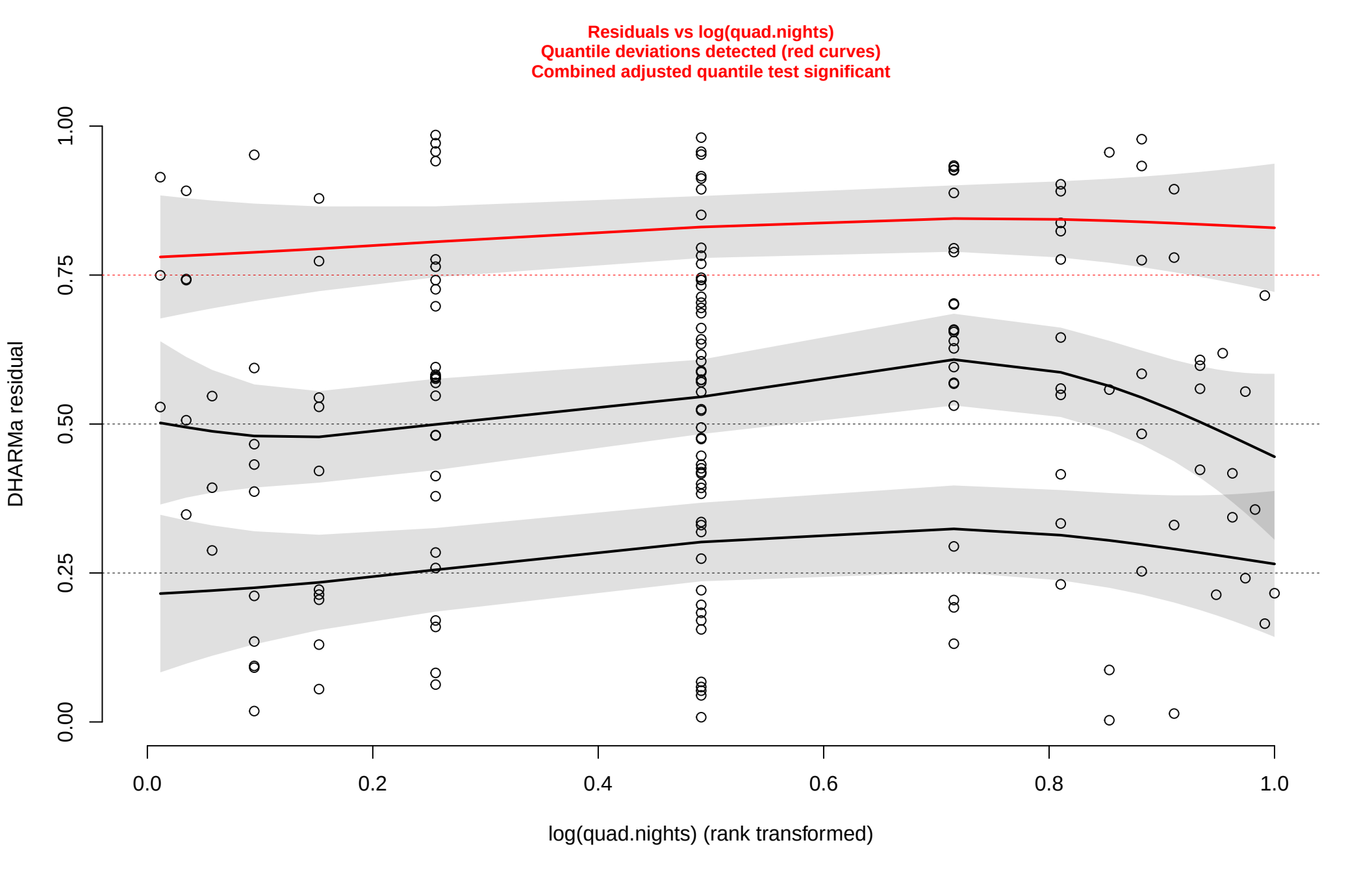
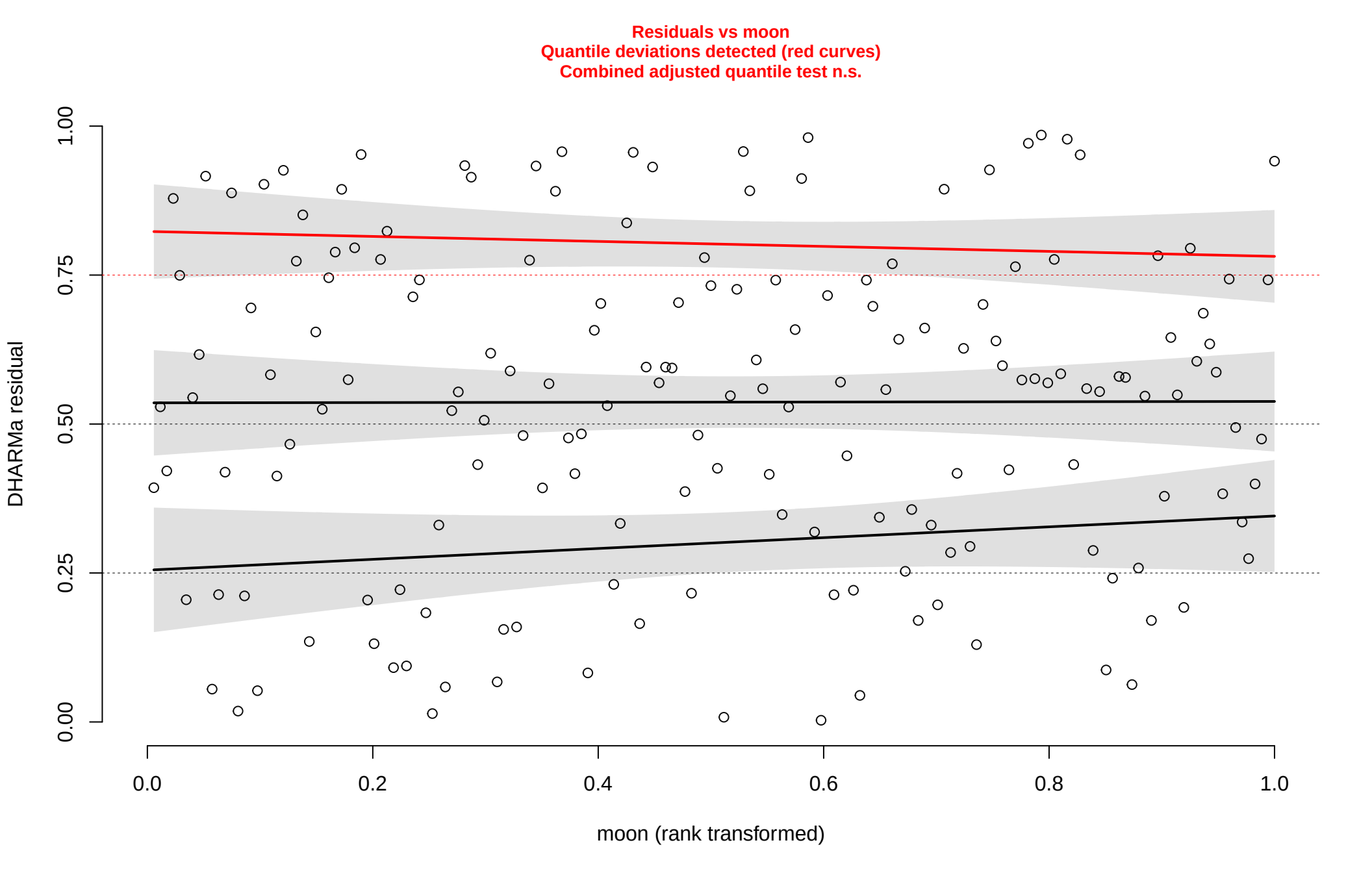
DHARMa residuals: Northern Region_LACI

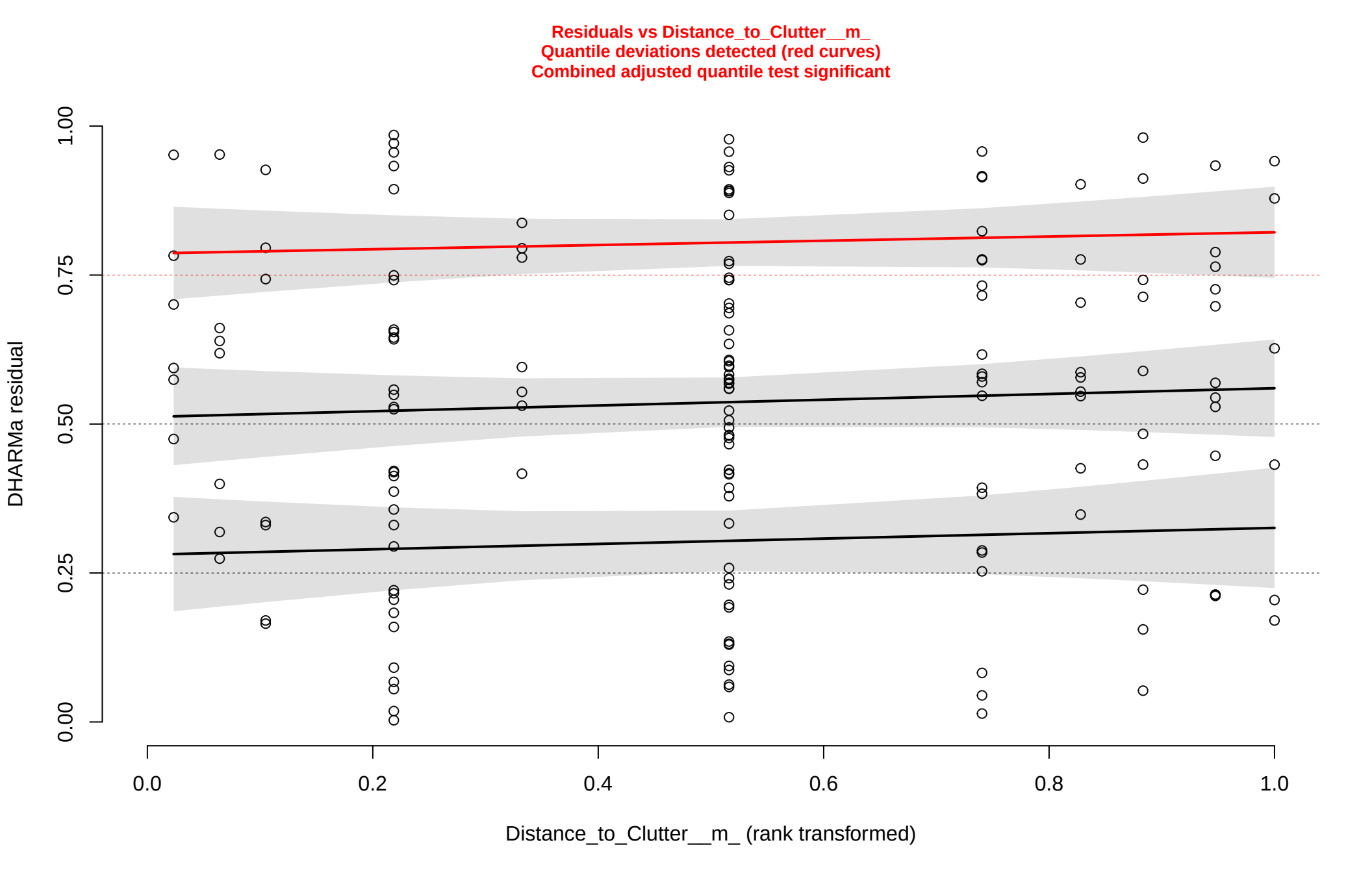


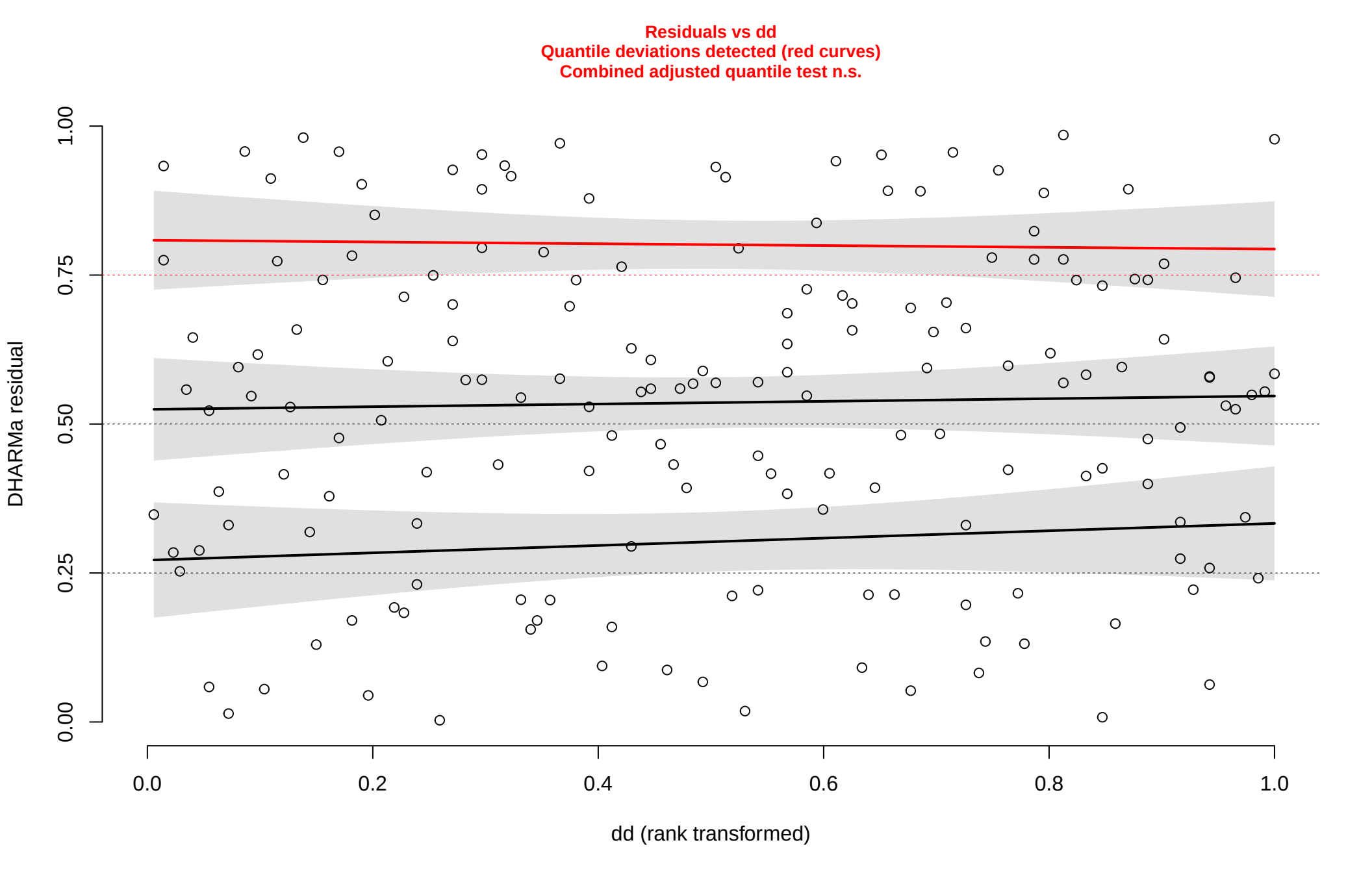


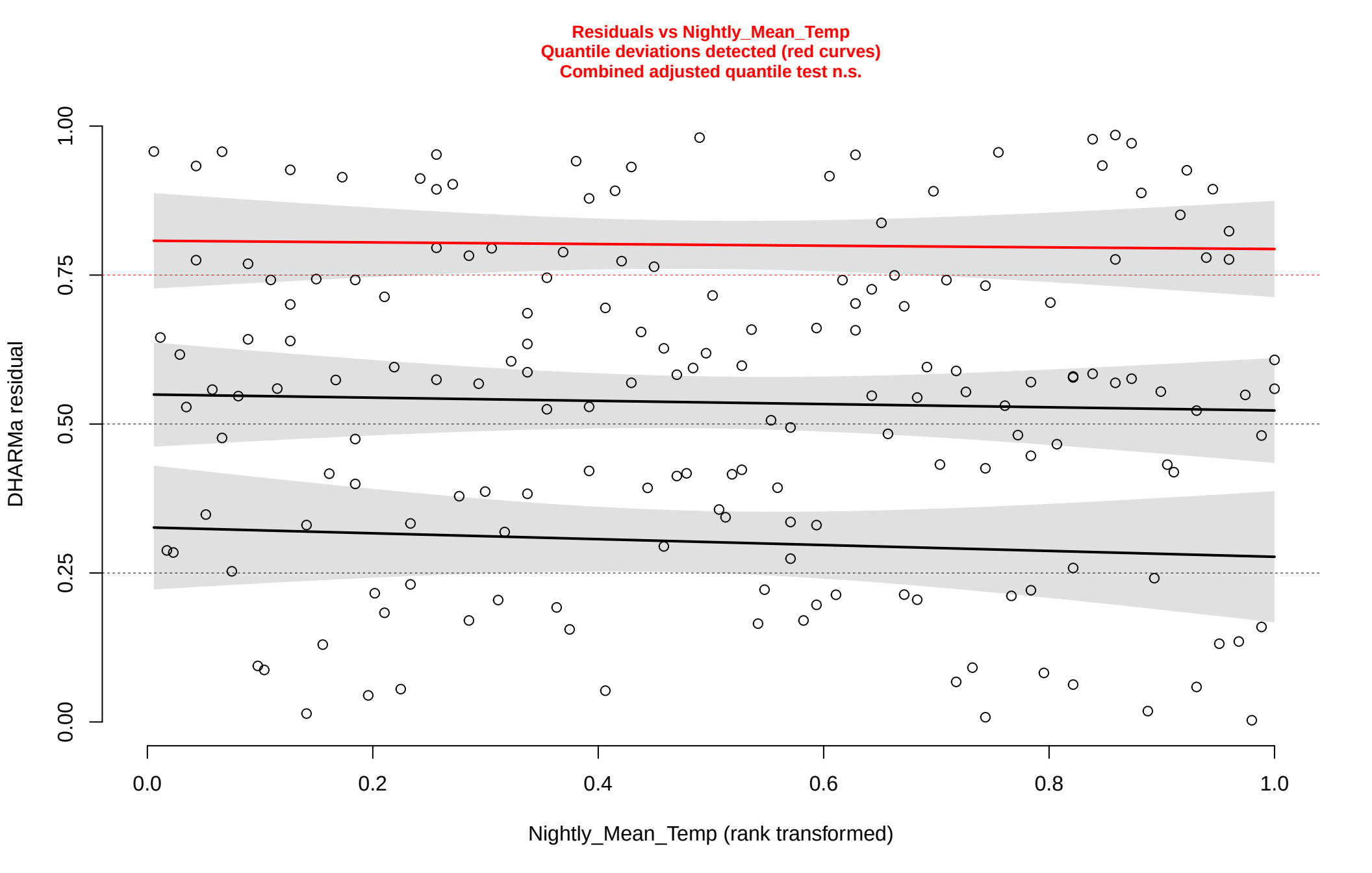

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

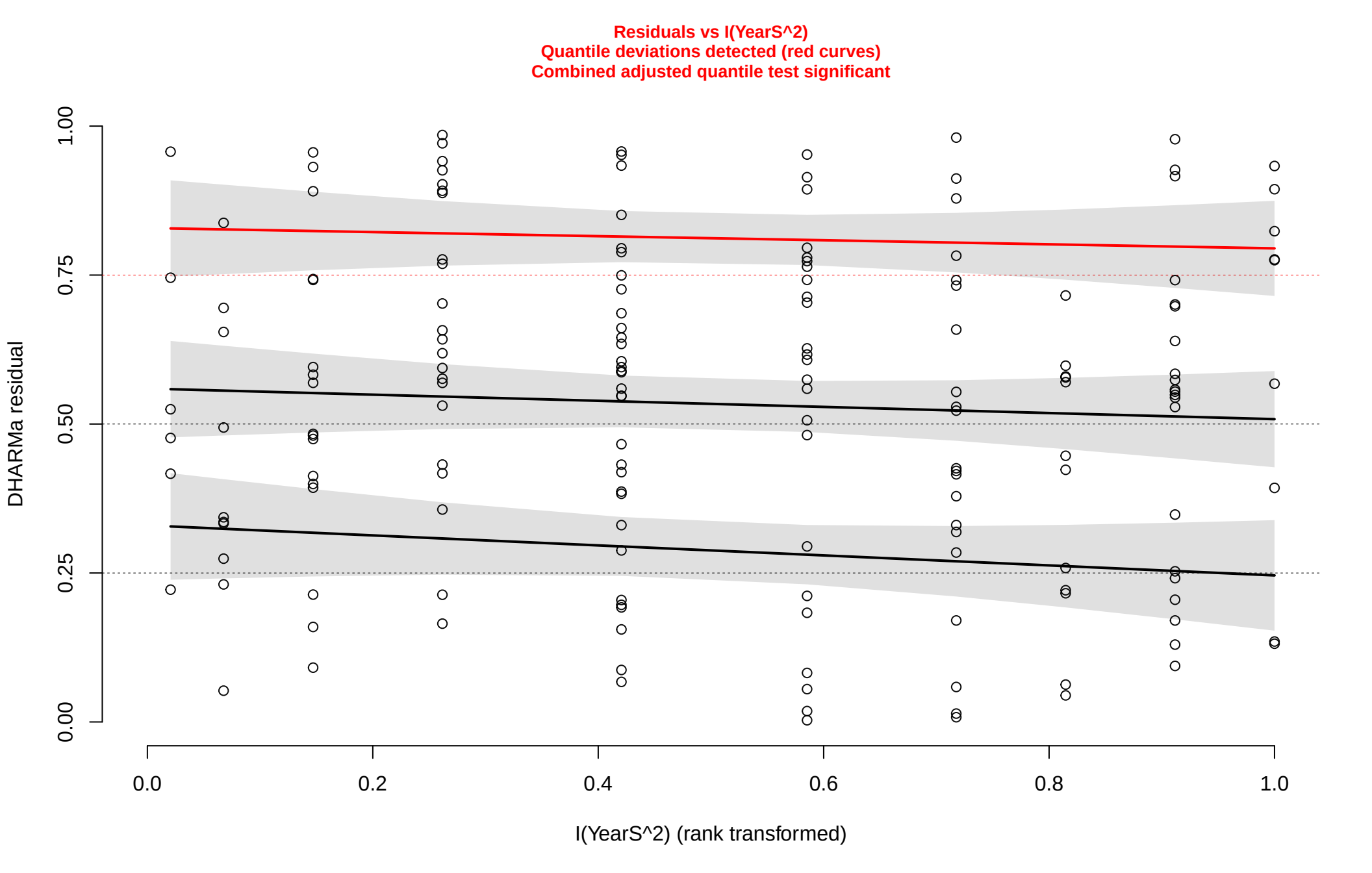

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

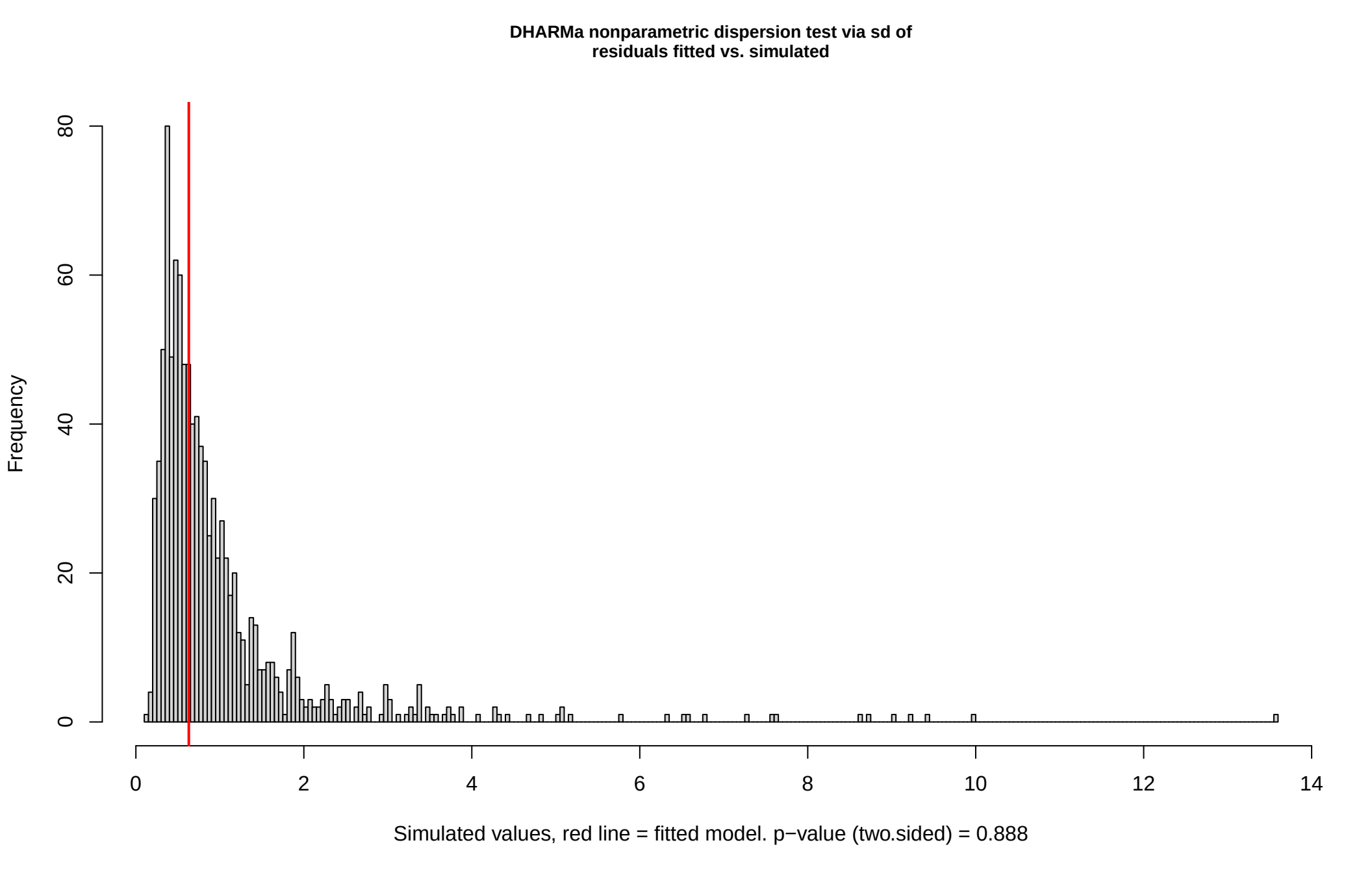

Residuals vs moon
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
moon (rank transformed)

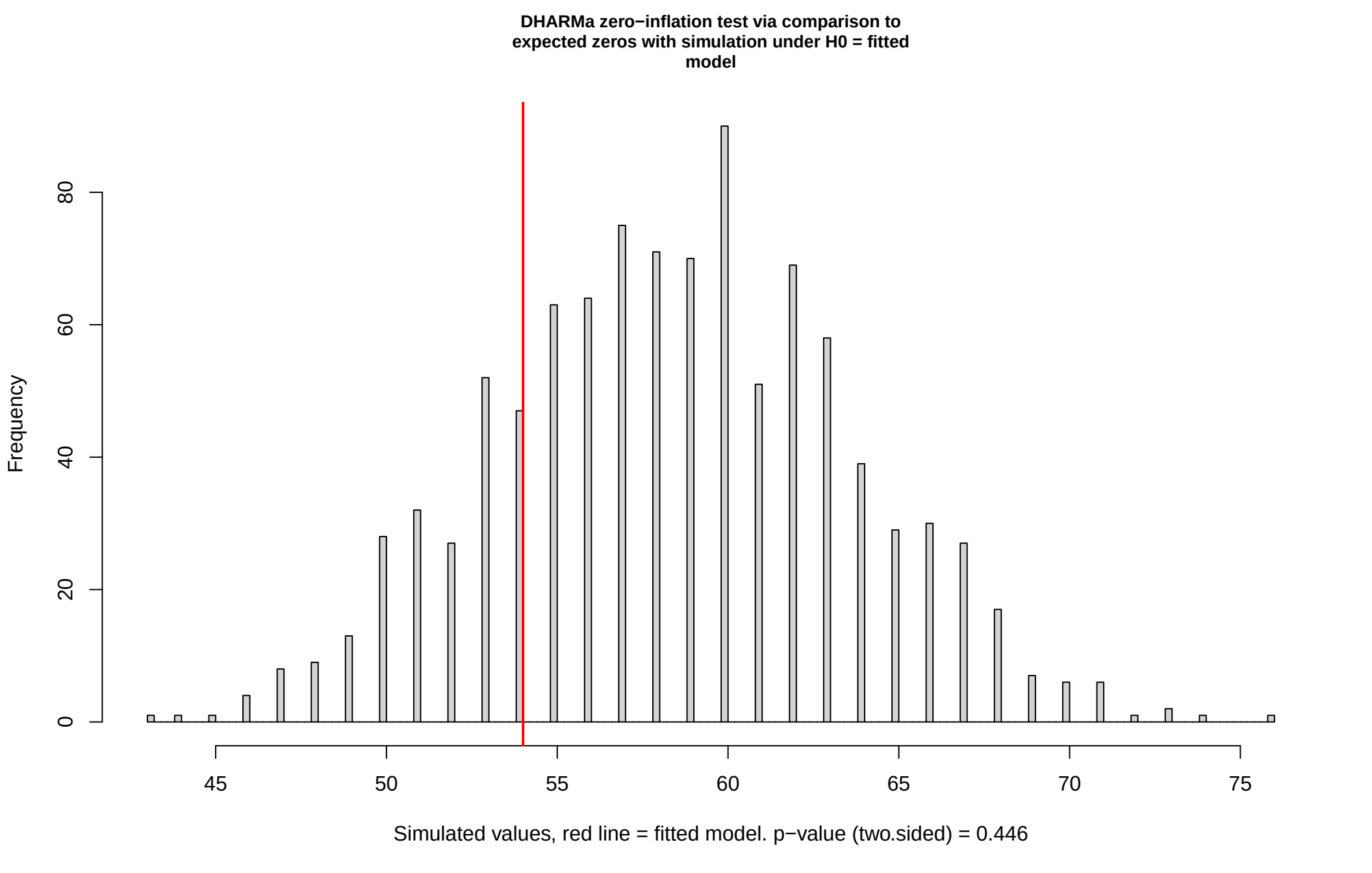

Residuals vs Distance_to_Clutter__m_
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

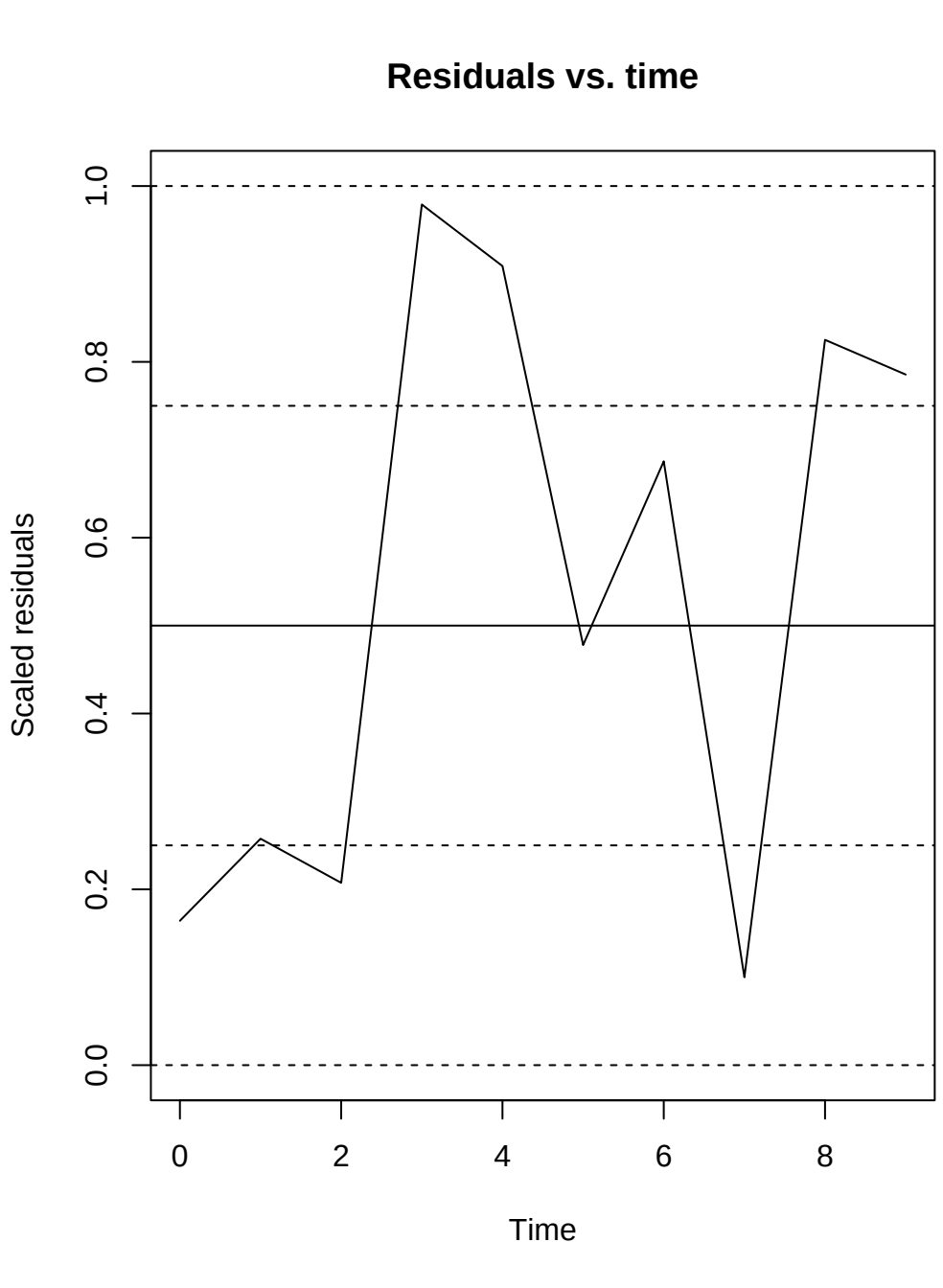

Residuals vs dd
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
dd (rank transformed)

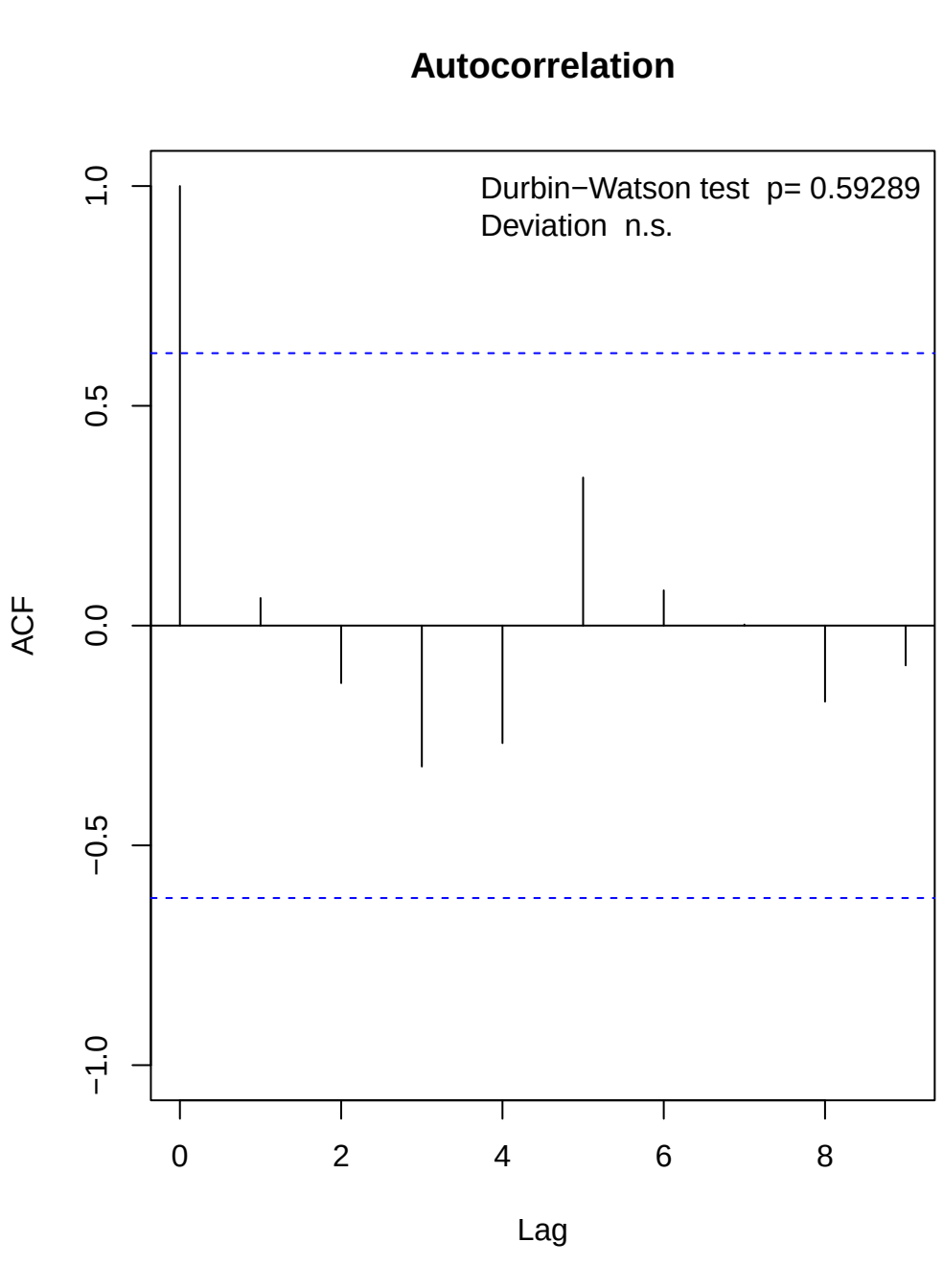

Residuals vs Nightly_Mean_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

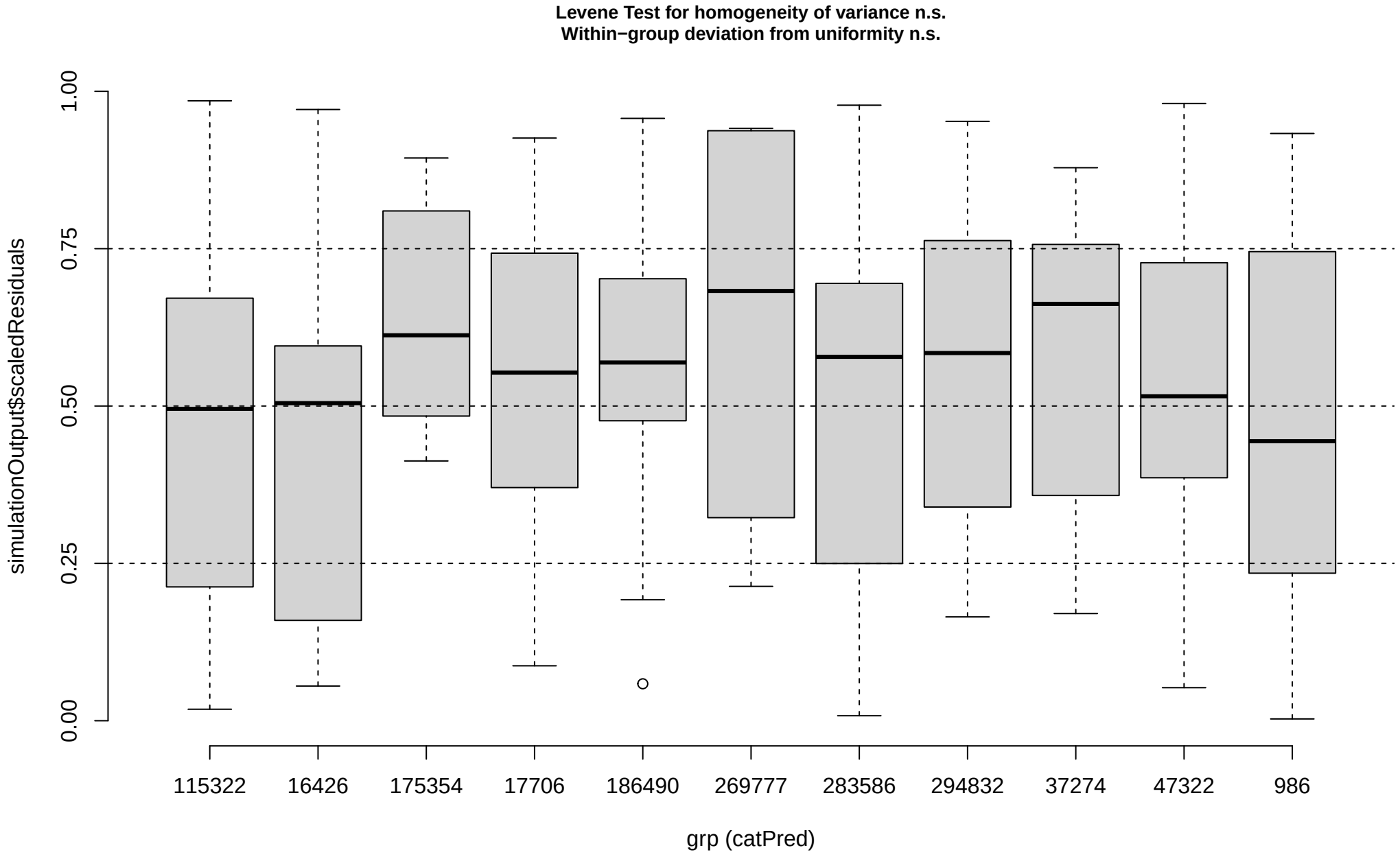

Residuals vs I(YearS^2)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
I(YearS^2) (rank transformed)

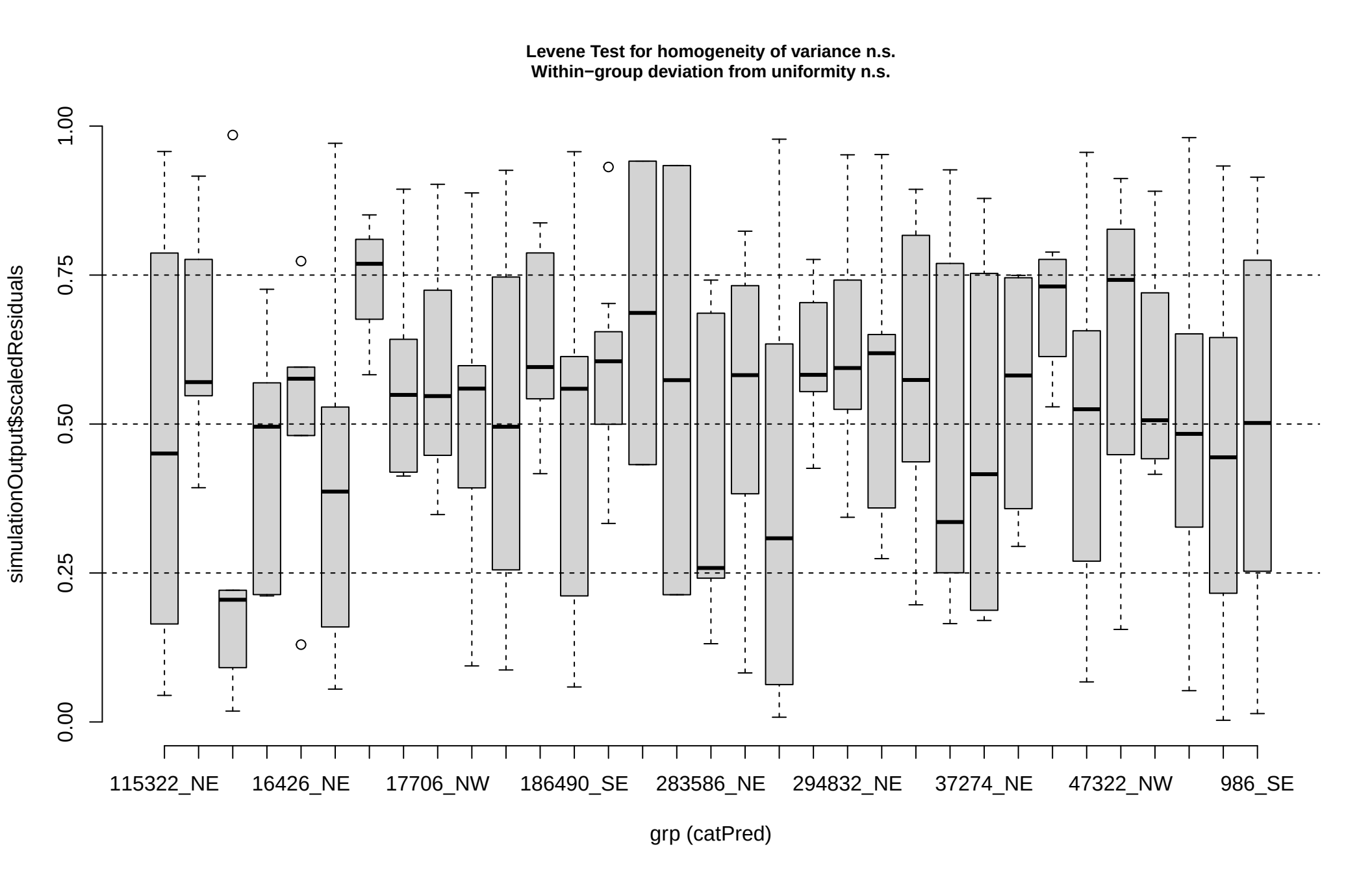

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
20
40
60
80
0
2
4
6
8
10
12
14
Simulated values, red line = fitted model. p−value (two.sided) = 0.888

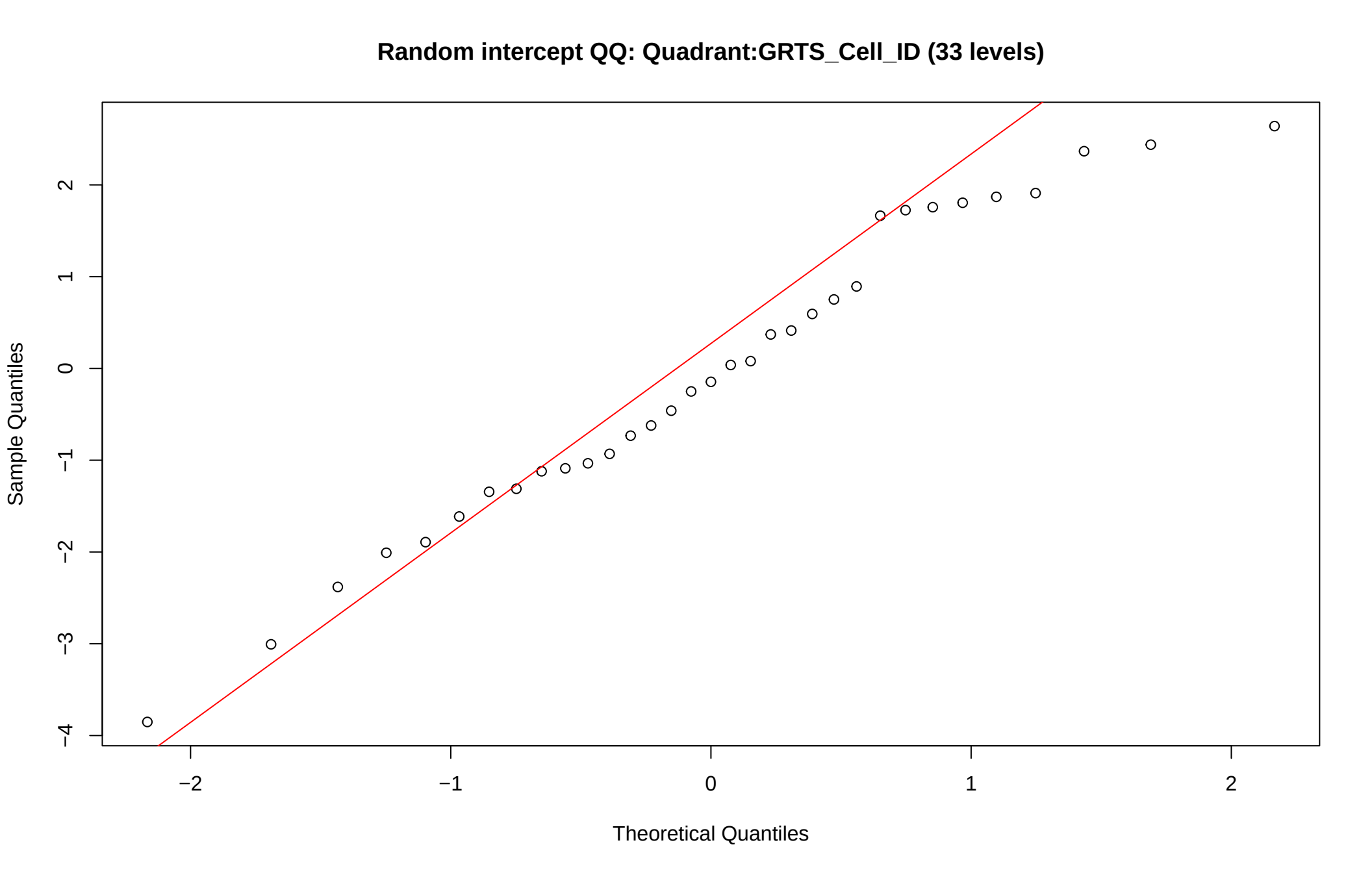

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
20
40
60
80
45
50
55
60
65
70
75
Simulated values, red line = fitted model. p−value (two.sided) = 0.446

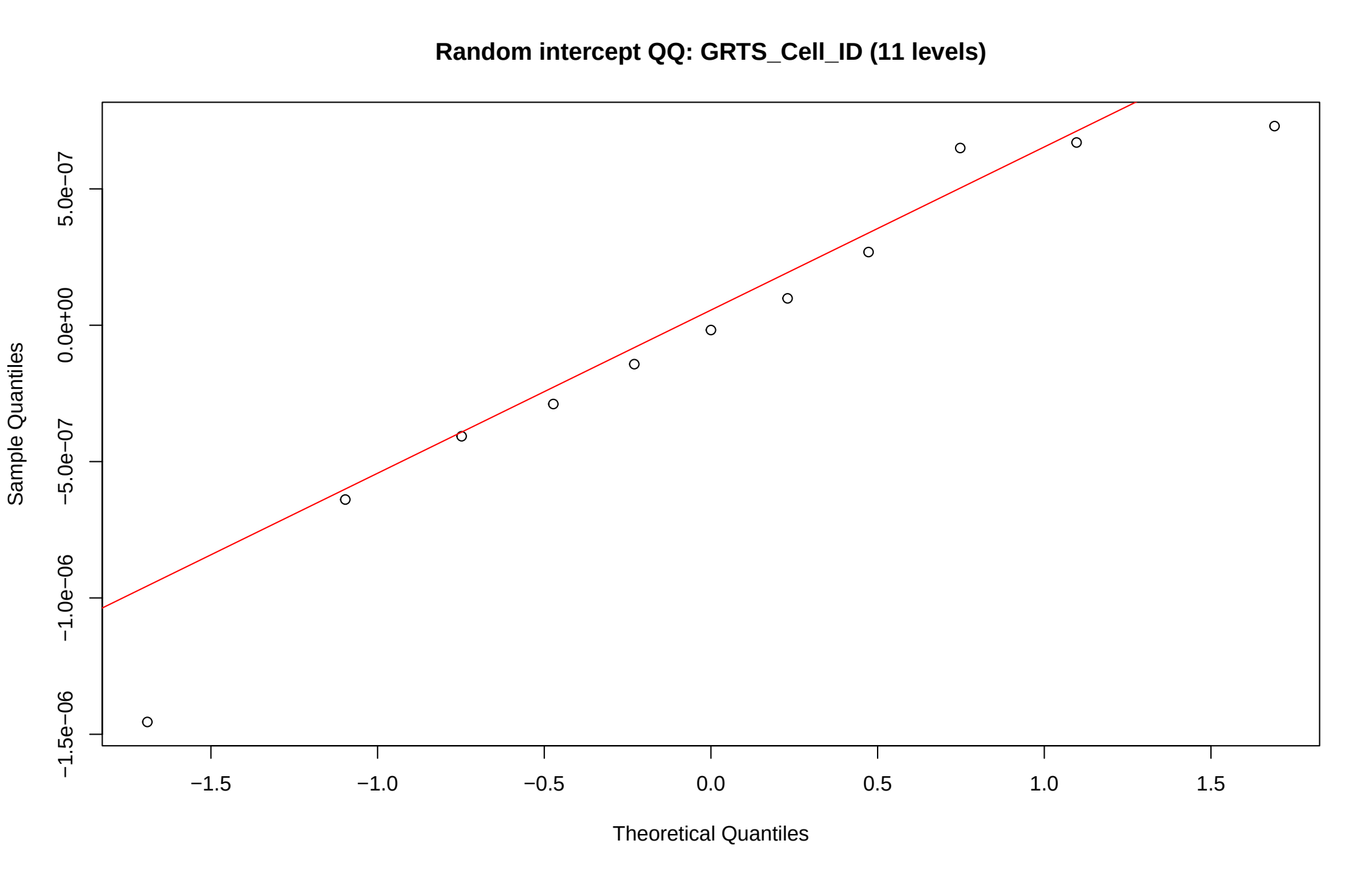

Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time


Autocorrelation
Durbin−Watson test p= 0.59289
Deviation n.s.
ACF
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
115322
16426
175354
17706
186490
269777
283586
294832
37274
47322
986
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
115322_NE
16426_NE
17706_NW
186490_SE
283586_NE
294832_NE
37274_NE
47322_NW
986_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (33 levels)
Sample Quantiles
Theoretical Quantiles
−4
−3
−2
−1
0
1
2
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (11 levels)
Sample Quantiles
Theoretical Quantiles
−1.5e−06
−1.0e−06
−5.0e−07
0.0e+00
5.0e−07
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5